Agenda

1) Continue with housing price prediction
   ↳ recap of most topics in the process

2) Answer a few questions asked before

✓ Q) Isolation forests → random splits or not ?

✓ Q) When can we have duplicate eigen values for a co-variance matrix

✓ Q) Why remove correlated features before doing PCA ?

Q) Quick example of eigen-vector finding

✓ Q) Does PCA help in feature selection

Q) Why eigen vectors are unit vectors ?

$x_1 = 3x_2$

$x_2 = \frac{1}{3} x_1$

$C_1 = 1x_1 + 2x_2 + 3x_3 \rightarrow (1, 2, 3) \rightarrow$ co-eff

$x_1 = 3x_2$

$C_1 = 1(3x_2) + 2x_2 + 3x_3$

$\rightarrow 0x_1 + 5x_2 + 3x_3 \rightarrow (0, 5, 3)$

$C_1 = 1x_1 + 2(\frac{1}{3}x_1) + 3x_3$

$= \frac{5}{3}x_1 + 0x_2 + 3x_3 \rightarrow (\frac{5}{3}, 0, 3)$

$(1, 2, 3) \longleftrightarrow (0, 5, 3)$

$\longleftrightarrow (\frac{5}{3}, 0, 3)$

multiple solutions possible

no unique soln

No reproducibility

$$\begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{bmatrix}$$

$\sigma_{12} = 0$

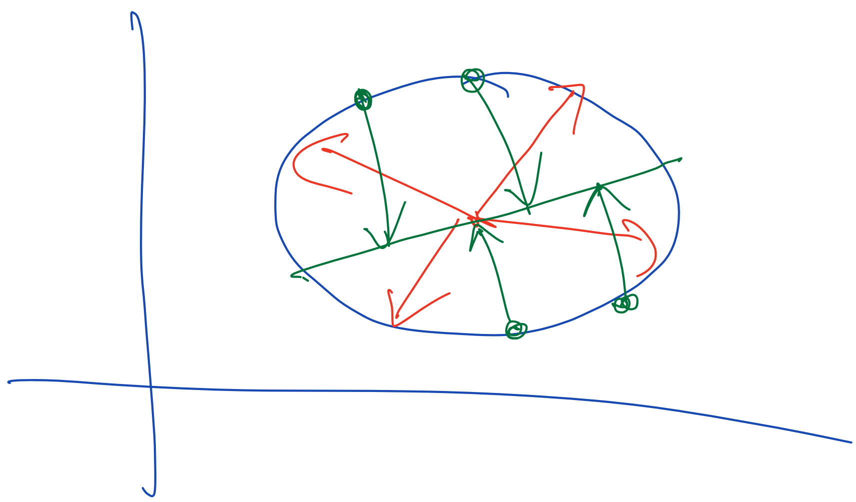
$\sigma_{11} = \sigma_{22} = v$ (say)

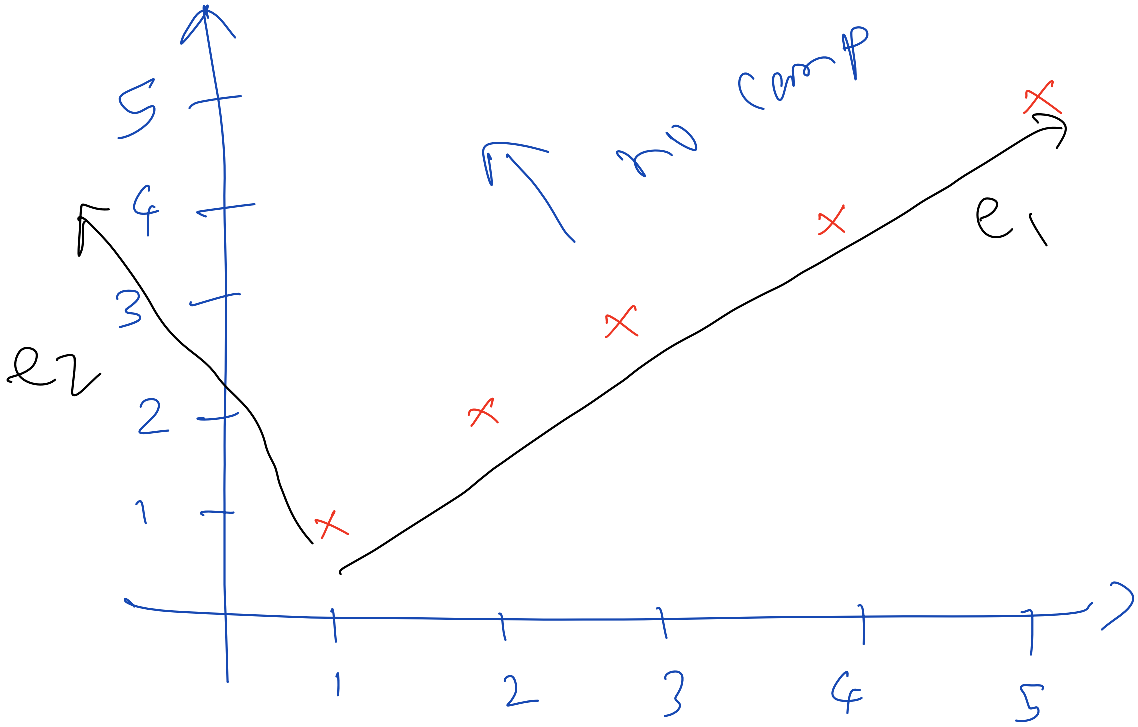
$$Cov = \begin{bmatrix} v & 0 \\ 0 & v \end{bmatrix}$$

$$\begin{bmatrix} v & 0 \\ 0 & v \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = v \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\left.\begin{array}{l} v u_1 = v u_1 \\ v u_2 = v u_2 \end{array}\right] \rightarrow u_1 \text{ \& } u_2 \text{ are uncorrelated}$$

u_1





$y = x$

cov =

$\lambda_1 = +ve$

$$\boxed{\lambda_2 = 0}$$